LearnWise AI: Project Documentation

Track D: Personalized Tutoring & Adaptive Learning

1. System Architecture & Tech Stack

LearnWise AI is built on a modular, privacy-first, local architecture designed for multimodal course ingestion and adaptive tutoring.

- **Frontend Application:** Streamlit (Python) providing a responsive, stateful interface with a unified dashboard, adaptive quiz generator, and grounded chat environment.
- **Language Model:** Local execution via Ollama utilizing `llama3.2` for zero-latency, private, and secure generation of explanations, adaptive questions, and semantic grading.
- **Vector Retrieval Pipeline:** TF-IDF vectorization with Cosine Similarity, specifically tuned for academic text matching and keyword overlap filtering.
- **Multimodal Ingestion Extractors:** `PyMuPDF` for PDF text, page preservation, and diagram extraction; `python-pptx` for slide and table ingestion; custom regex-based parsing for video transcripts (.vtt, .srt) with timestamp mapping.

2. Grounding & Retrieval Methodology

To prevent hallucinations and guarantee academic trustworthiness, LearnWise AI implements a strict Source-Grounded retrieval framework:

Multimodal Indexing

Uploaded materials are automatically chunked into overlapping 3-sentence blocks. Crucially, every chunk retains a hard link to its origin (exact page number, slide index, or video timestamp). Content is automatically classified into subtopics using term-frequency analysis.

Threshold-Based Refusal

When a user queries the AI, the system requires a minimum cosine similarity score (>0.08) and lexical overlap threshold before generating an answer. If the threshold is not met, the system strictly declines the query or explicitly flags it as "Off-Material / Supplemental AI Knowledge", guaranteeing separation of facts from external LLM knowledge.

Inline Citation Generation

The LLM is instruction-prompted to assemble its explanations using only the retrieved blocks. Every generated claim includes a mandatory inline bracket citation (e.g., `[Page 12]` or `[Timestamp 04:30]`) linking back to the visual source context in the UI.

3. Learner Model Approach

Personalization is driven by **Bayesian Knowledge Tracing (BKT)**, which maintains a probabilistic estimate of mastery ($P(L)$) for each discovered topic.

- **Intake Calibration (Cold-Start):** New users select their experience level, initializing the prior probability $P(L_0)$ (Beginner: 0.15, Intermediate: 0.40, Advanced: 0.70).
- **Probabilistic Updates:** After every adaptive question, $P(L)$ is updated using standard BKT parameters: Slip ($P(S) = 0.10$), Guess ($P(G) = 0.20$ for MCQ, 0.05 for Open), and Transition ($P(T) = 0.15$).
- **Conversational Reinforcement:** The model applies a mild transition reinforcement when a student actively queries a topic in the AI Tutor chat, reflecting active learning outside of formal assessments.